

Research Interest	Developing efficient methods for large language model pre-training, with a focus on training dynamics, hyperparameter transfer, and optimizers for large-scale training.	
Education	University of Southern California <i>Doctor of Philosophy, Computer Science</i>	Expected 2027.12
	University of Southern California <i>Master of Arts, Pure Mathematics</i>	Expected 2027.5
	New York University <i>Master of Science, Computer Science</i>	2021.5
	University of Michigan, Ann Arbor <i>Bachelor of Science in Engineering, Electrical Engineering</i>	2019.5
Experience	Honeywell <i>AI PhD Intern</i> Developed AI models to solve targeted business problems.	2026.5 - 2026.8
	Huawei <i>Machine Learning Engineer</i> Worked on object detection in construction sites.	2021.9 - 2022.5
Publications	(* indicates equal contribution.) 7. Towards Self-Adaptive Learning: Continual Learning under Harsh Conditions. <i>Roozbeh Razavi-Far, Ehsan Hallaji1, Alireza Fathalizadeh, <u>Mengxi Wu</u>, Mohammad Rostami.</i> Submitted 6. Gecko: An Efficient Neural Architecture Inherently Processing Sequences with Arbitrary Lengths. <i>Xuezhe Ma*, Shicheng Wen*, Linghao Jin*, Bilge Acun*, Ruihang Lai*, Bohan Hou, Will Lin, Hao Zhang, Songlin Yang, Ryan Lee, <u>Mengxi Wu</u>, Jonathan May, Luke Zettlemoyer, Carole-Jean Wu.</i> Submitted 5. GQA-μP: The Maximal Parameterization Update for Grouped Query Attention and Fully Sharded Data Parallel. <i>Kyle R. Chickering*, Huijuan Wang*, <u>Mengxi Wu</u>*, Alexander Moreno, Muhao Chen, Xuezhe Ma, Daria Soboleva, Joel Hestness, Zhengzhong Liu, Eric P. Xing.</i> Submitted 4. Curvature Diversity-Driven Deformation and Domain Alignment for Point Cloud. <i><u>Mengxi Wu</u>, Hao Huang, Yi Fang, Mohammad Rostami.</i> Transactions on Machine Learning Research 2025 3. Graph Harmony: Denoising and Nuclear-Norm Wasserstein Adaptation for Enhanced Domain Transfer in Graph-Structured Data. <i><u>Mengxi Wu</u>, Mohammad Rostami.</i> Transactions on Machine Learning Research 2024	

2. **Streaming Approach to In Situ Selection of Key Time Steps for Time-Varying Volume Data.**

Mengxi Wu, Yi-Jen Chiang, Christopher Musco.

[Eurographics/IEEE Conference on Visualization 2022](#)

1. **3D Point Cloud Completion with Geometric-Aware Adversarial Augmentation.**

Mengxi Wu, Hao Huang, Yi Fang.

[International Conference on Pattern Recognition 2022](#)

Teaching

University of Southern California

CSCI 544 Applied Natural Language Processing, *Teaching Assistant*

CSCI 570 Analysis of Algorithms, *Teaching Assistant*

New York University

ECE-GY 9123 Deep Learning, *Teaching Assistant*

Services

Conference Reviewing

EMNLP, ICLR, ICML, NeurIPS

Journal Reviewing

Transactions on Machine Learning Research

Skills

Programming Languages

C/C++, Python, Java, R, MATLAB, Swift

Libraries and Tools

PyTorch, Tensorflow, PySpark, Hadoop